1 Six centers



Center 0
12,413 images



Center 1
3,954 images



Center 2
3,363 images



Center 3
2,259 images



Center 4
819 images



Center 5
439 images

2 Two split types

Image-level
siblings cross the split



train test

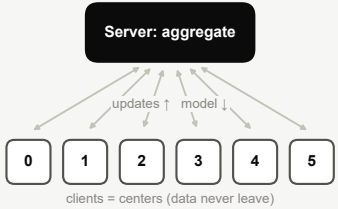
Lesion-grouped
lesions kept whole



train test

5 random re-splits each

3 Federated training



Encoders

A	EfficientNet-B0, fine-tuned
A-frozen	EfficientNet-B0, frozen
B	PanDerm-Base, frozen

Methods

FedAvg · FedProx · SCAFFOLD · FedBN
FedAvg-FT · FedPer · pooled · local-only

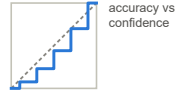
4 Reliability evaluation

RQ2
Leakage inflation



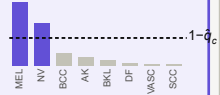
full minus new-lesion BA

RQ1
Calibration



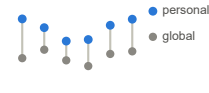
ECE per center

RQ1 · RQ4
Conformal coverage



EB-ShrinkFCP (proposed)

RQ3
Personalization



H3b; per-center change

Stress tests leave-one-center-out models · PAD-UFES-20 smartphone images (domain shift)

Pre-registered protocol v0.11 → tagged code → runs → pre-specified tests